



CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

List Of Practical's (Indicative & Not Limited To)

- ## Searching

- ## Stack

- ## Queue

- ## Linked List

21. Create a single linked list and perform basic operations (insertion, deletion, traversal).
22. Create a double linked list and perform basic operations (insertion, deletion, traversal).
23. Create a circular linked list and perform basic operations (insertion, deletion, traversal).
24. Reverse a single linked list.
25. Check if a linked list is palindrome.
26. Reverse a double linked list.

27. Find the middle element of a single linked list.

28. Find the middle element of a double linked list.

29. Merge two sorted single linked lists.

30. Detect and remove a loop in a circular linked list.

Binary Tree

31. Construct a code to Insert, Delete and search and update a data in Binary Search Tree (BST)

32. Construct a code for Tree Traversal (Preorder, Inorder, Postorder).

33. Construct a code Count the number of Leaves in a Binary Tree

34. Construct a code to find the Height of a Binary Tree

35. Construct a code to print all Paths from the Root to Leaf Nodes in a Binary Tree

36. Construct a code to convert a Binary Tree to its Mirror Tree

BST

37. Construct a code to find the Node with Minimum Value in a Binary Search Tree.

38. Construct a code for Binary Search Tree (BST) Implementation.

39. A program to check if a Binary Tree is a Binary Search Tree (BST)

AVL Tree

40. Construct a code to check if a Binary Tree is a Balanced Binary Tree

Graph

41. Construct a code to represent graph using adjacency matrix and adjacency list.

42. Implement BFS and DFS algorithm.

43. Implement the minimum cost spanning tree.

Sorting

44. Implement bubble sort in a non-recursive way.

45. Implement selection sort in a non-recursive way.

46. Implement insertion sort in a non-recursive way.

47. Implement Merge sort in a non-recursive way.

48. Implement Merge sort in a recursive way.

49. Implement Quick sort in a recursive way.

50. Implement Heap sort in a non-recursive way

Total Hours: 48 hrs.